

Particulate Matter Monitoring & Health Effects

Issue 2026-09-03 | Entry window 3 September 2026 (single day) | 38 records in scope, 88 screened out

38

RECORDS IN SCOPE
(88 SCREENED OUT)

9

SUBTOPIC CLUSTERS
OF TEN OCCUPIED

12

EFFECT ESTIMATES
(FOUR SOURCES)

11

SENSING /
INSTRUMENTATION

23

RECORDS WITH NO
HEALTH ENDPOINT

Signal of the day

Plantower's new PMSX003N is better than the PMS5003T at dust and worse at humidity — and the two failure modes are the same failure mode.

Kaur, Mangin and Kelly (*Aerosol Sci. Technol.*) co-located three PMSX003N, three PMS5003T and three OPC-N3 against FEM monitors at a Utah DEQ site for five months (Dec 2025–Apr 2026), through dust events, winter inversions, freezing fog, mist, snow and rain. During dust the PMSX003N tracked the reference where its predecessor did not ($PM_{2.5}$ slope 1.85–1.91, $R^2 > 0.7$ against slope 0.15, $R^2 < 0.37$ for the PMS5003T; PM_{10} slope 0.91–1.07 against ~0.02), and under high humidity it **substantially overestimated both fractions**. Laboratory size-binning explains both: the sensor **preferentially assigns particles** $>1\ \mu m$ to the 2.5–5 μm bin. Coarse mineral dust is thereby recovered rather than lost, and hygroscopically grown droplets are recruited into the same oversized bin. The design change that fixes coarse-PM response is the design change that makes RH correction mandatory — so a humidity-uncorrected PMSX003N network is not a drop-in upgrade over a PMS5003T network, it is a different bias. Note the $PM_{2.5}$ slope: even at its best the sensor reads roughly 1.9× the reference during dust.

What changed today

- **The largest single-day issue this watch has shipped, and it is instrument-led.** 38 records against a previous single-day maximum of 36 (27 August); only the retrospective July back-scan is bigger. **Eleven** sit in *Sensing, forecasting & instrumentation* and **twelve** in *Exposure assessment & modelling*: 23 of 38 before a single health endpoint. Volume is a one-day artefact of a fat *ES&T* by-ISSN deposit, but the sensing half is substantive rather than incidental.
- **Two independent low-cost-sensor papers, and they disagree about what LCS are for.** Kaur et al. treat the sensor as an instrument to be characterised against a reference; Guth et al. (*Atmosphere*) treat it as the only practical way to get *time-resolved* emission factors across **19 prescribed burns**, where CO EFs swing +3260% with combustion phase. The second use case tolerates a bias the first would reject, because within-burn contrast is the measurand.
- **Component-resolved epidemiology produced two effect sizes that cannot both be read the same way.** Chen K et al. report per-IQR frailty HRs of 1.142 (1.055–1.236) for organic matter and 1.131 (1.041–1.229) for black carbon in CHARLS ($n=9,603$) — component ordering above bulk $PM_{2.5}$ (1.100), and biologically legible. Chen Y et al. report **OR 2.23 (2.19–2.27)** for sulfate on congenital heart disease across **2.6 million** Guangdong births, with the effect peaking in *pregnancy week 1*. A doubling of CHD odds with a CI of width 0.08 is not a plausible causal contrast; it is what a spatially smooth satellite exposure does against a very large register. Cite the first, quarantine the second.
- **A perspective worth reading against your own measurement plan.** Gao et al. (*ES&T*) argue that the recurring *inverse* O_3 –health associations are an artefact of measuring O_3 as a proxy for an unmeasured oxidant field: low O_3 can mark active oxidative chemistry producing toxic secondary products nobody instruments for. The same argument applies verbatim to PM mass as a proxy for particle oxidative potential.
- **Radon methodology, transferable wholesale.** Giang et al. co-located **45 consumer radon monitors (15 models × 3 units)** against a reference for 98 days in an occupied home: 8 of 15 models within $\pm 25\%$ relative percent error, worst-case **–47% to +220%**, intra-model CV mostly under 15%. Different analyte, identical QA problem to a consumer PM network — and a rare published estimate of *between-model* spread at consumer price points.
- **Africa, twice, and both times as a gap statement.** Wright et al. find only 13 African case-control studies in a decade (indoor $PM_{2.5}$ 96–177 $\mu g/m^3$, ambient to 259) and **none** examining pollution–heat interaction; Hulme et al. link home and school exposures for 919 Accra schoolchildren. A third African record (Lagos speciation, Hopke a co-author) ships on metadata only.
- **Nil legs.** Consensus nil for the fifth run; arXiv answered without a 429 but its newest submission is **2026-08-20**; OpenAlex 429; trial registry a true zero.

Corpus analytics

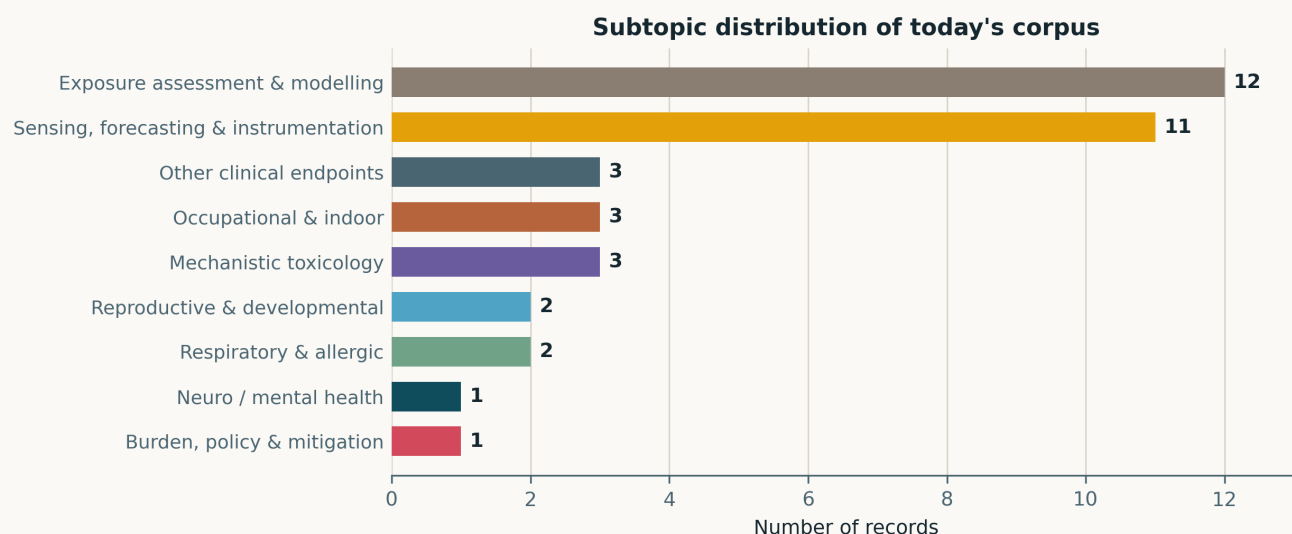


Fig. Subtopic assignment of the 38 in-scope records, single-label by dominant contribution. *Exposure assessment & modelling* (12) and *Sensing, forecasting & instrumentation* (11) hold 61% of the issue between them. Only *Cardiovascular & metabolic* is vacant; nine or more clusters have been occupied on ten previous issues, most recently all ten on 27 August. The single *Burden, policy & mitigation* record is the China 2035 NDC co-benefit model, and the single *Neuro / mental health* record is an $n=44$ exploratory PCA, so two of those nine cells are thin.

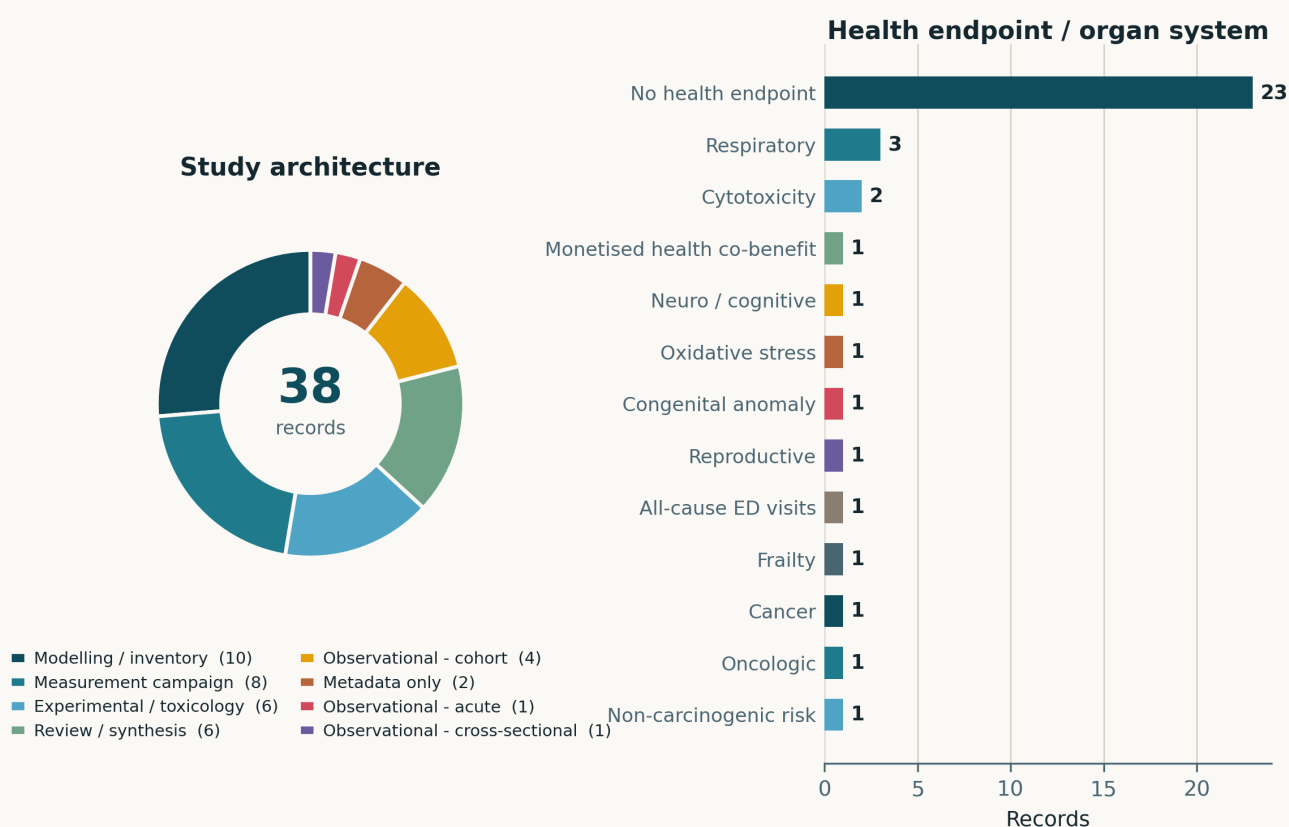


Fig. Left — study architecture over 23 distinct design labels, 14 of which are new to the design map and were added to `plots.py` in the same edit that wrote them. *Modelling / inventory* (10) and *Measurement campaign* (8) lead; *Review / synthesis* holds 6 (three systematic reviews, three critical reviews). Only 2 records ship on *Metadata only*, against 6 on 2 September. Right — 23 of 38 (61%) carry no human health endpoint. That is the largest count the watch has recorded but not the largest share: 20 August reached 72% at $n=18$ and 2 August 71% at $n=31$, and the last three issues have run 58%, 60% and 61%, so the level is a plateau rather than a jump. Read *Cancer* and *Oncologic* as two different quantities that the canonical map does not merge: the former is 1,625 observed incident cancers in a Canadian birth cohort, the latter a computed carcinogenic-risk index in a metadata-only Lagos record.

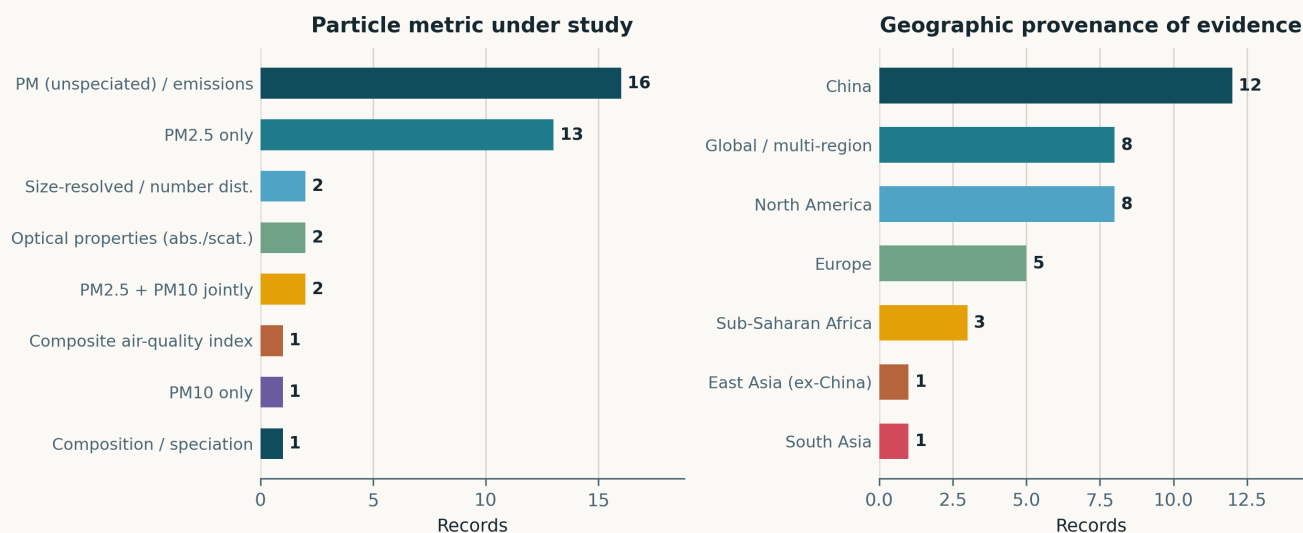


Fig. Left — particle metric. *PM (unspeciated) / emissions* at 16 is inflated by the instrument and chemistry records whose measurand is not a regulated mass fraction (lidar smoke mass, CCN, INP, organosulfates, filtration efficiency, and the one gas-phase NO₂ sensor carried here). *PM_{2.5} only* at 13 is the honest epidemiological core. Right — geography. China 12, North America 8, Europe 5. The *Global / multi-region* bar of 8 is **not** eight multi-region studies: four are genuinely multi-region reviews and four are laboratory records, which this pipeline routes to that bucket by intent. **Sub-Saharan Africa at 3 ties the watch record**, set on 6 August; South Asia (Delhi) and East Asia ex-China (South Korea) contribute one each; Latin America, the Middle East, Southeast Asia and Oceania are absent.

Life-course windows addressed today (records may span several stages)

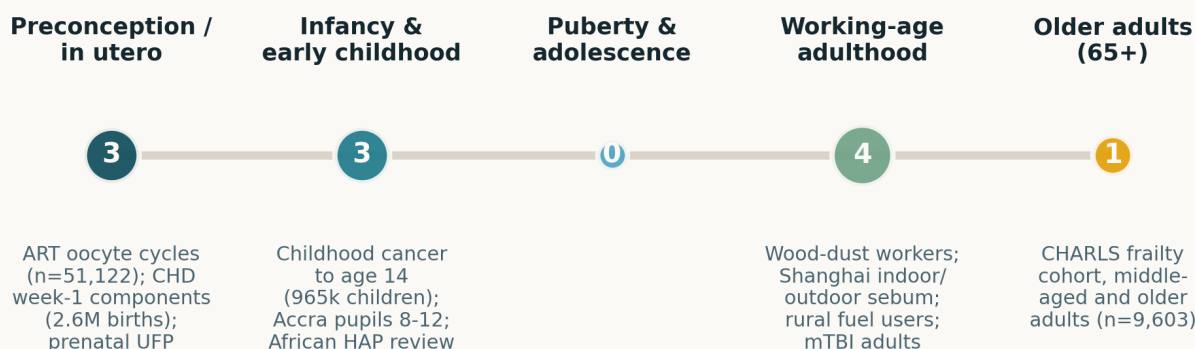


Fig. Life-course windows addressed; counts are non-exclusive and cover only the 15 records with a human endpoint. **Puberty and adolescence is empty** for the second consecutive issue. Working-age adulthood (4) is the largest cell, carried by two occupational records, the rural solid-fuel personal-monitoring study and the mTBI cohort; it also led on 29, 30 and 31 August, so this is the resumption of a pattern rather than a new one.

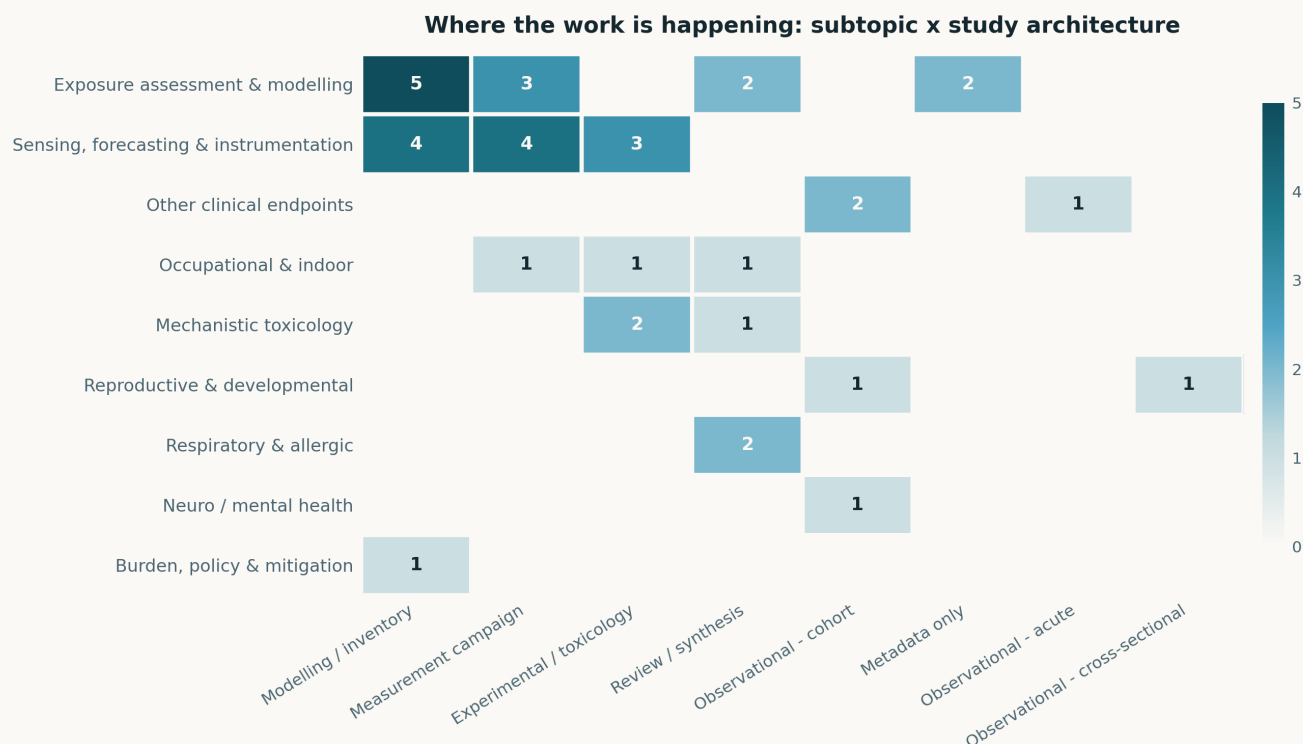


Fig. Subtopic × study-architecture cross-tabulation. The dense cell is *Exposure assessment & modelling* × *Modelling / inventory*; the *Sensing* row spreads across measurement campaign, modelling and laboratory development rather than concentrating, which is what a genuinely instrument-heavy day looks like as opposed to a day of forecasting papers filed under sensing.



Fig. Twelve estimates from **four** sources, and they are not commensurable. Six are per-IQR component-specific frailty hazard ratios from one CHARLS analysis and are strongly correlated with one another (the components co-vary in the same TAP exposure surface). Two are per-10,000 cm⁻³ prenatal-UFP cancer hazard ratios whose intervals both cross the null. Two are per-IQR ozone rate ratios on 3.48 million ED visits — near-null point estimates with intervals so tight they render as bare markers, which is the correct visual for a 1.9% excess. The two outliers at the top of the panel are the Guangdong CHD odds ratios; their intervals are narrower than any biological reading justifies and they should be treated as a register-scale artefact, not as the day's largest true effect. Wood-dust spirometry (pooled MD −8.57 %pred FEV₁) is absent because a mean difference has no place on a log-ratio axis.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Reproductive & developmental

2 records

Xu ZL et al. 2026 Biomed Environ Sci

Component-specific PM_{2.5} windows and oocyte yield in assisted reproduction

51,122 ART cycles with individual PM_{2.5} and its sulfate, nitrate, ammonium, organic-matter and black-carbon fractions assigned over recent (0–3 months), distal (4–12 months) and cumulative windows; negative binomial regression on total, mature and normally fertilised oocyte counts, DLNM for window identification, mixture models for joint effect. Higher exposure tracked poorer oocyte outcomes throughout, with **stronger effects for the distal window** and two sensitive periods — 1 month and 6–11 months before retrieval. Sulfate and ammonium dominated the mixture. The 6–11 month window is biologically coherent with primordial-to-antral follicle transit, which is the paper's strongest argument. Its weakness is that ART patients are a selected population and the exposure model is residential, so a distal window is also the window over which residential misclassification is largest.

doi: 10.3967/bes2026.072

Chen Y et al. 2026 Sci Rep

PM_{2.5} components in early pregnancy and congenital heart disease

2,601,300 live births and stillbirths, Guangdong 2014–2017, with satellite-derived component exposures resolved to the week of gestation and quantile g-computation for mixture weights. Effects peak in **pregnancy week 1**: sulfate **OR 2.23 (2.19–2.27)**, black carbon **1.95 (1.91–1.98)**, joint weights 86.8% and 13.2%. Read the intervals before the point estimates. A near-doubling of CHD odds with a 95% CI of width 0.08 implies a precision no exposure model at this spatial scale supports, and a week-1 peak in a satellite product with month-scale effective resolution is more consistent with seasonal and spatial confounding than with a window-specific teratogenic mechanism. The scale of the register is real; the causal contrast is not established.

doi: 10.1038/s41598-026-59940-7

Respiratory & allergic

2 records

Wright et al. 2026 Curr Environ Health Rep

Air pollution and respiratory outcomes in Africa — a review that mostly documents absence

PRISMA search of PubMed, ScienceDirect and Elicit for African case-control studies 2015–2025 returned **13 studies from 10 countries**. Reported concentrations are the finding: indoor PM_{2.5} **96–177 µg/m³** in biomass-using homes and ambient PM_{2.5} to **259 µg/m³**, an order of magnitude above WHO guidance. Nitrogen oxides tracked reduced lung function in children; household air pollution tracked under-five mortality and low birthweight; associations with acute respiratory infection were inconsistent across settings. **Only three studies measured temperature and none examined heat–pollution interaction**, which is the stated motivation for the review and which it therefore cannot deliver. Restricting to case-control designs also excludes the cohort and panel literature, so this is a floor on the African evidence base, not a census.

doi: 10.1007/s40572-026-00561-7

Cui L et al. 2026 Clin Rev Allergy Immunol

Epidemic thunderstorm asthma in China as an Artemisia phenotype

54 primary studies (43 international, 11 Chinese — Yulin, Hohhot, Chongqing, Lanzhou, Shenzhen) from 436 records, risk of bias by Newcastle–Ottawa and Murad, certainty by GRADE. Three patterns recur: Chinese events are associated with **mugwort (*Artemisia*)** rather than ryegrass sensitisation; **40–60% of Chinese cases and up to 78% of Australian and UK cases had no prior asthma diagnosis**; and individual Chinese localities report recurrence. No meta-analysis, and GRADE certainty for direct pollen rupture, pollution-mediated severity and pre-emptive treatment is **low to very low**. The measurement gap is explicit and is the actionable part: Chinese reports identify sensitisation only to genus level and **do not speciate the airborne pollen**, so no aerobiological trigger threshold can be set. The authors' first research priority is field aerosol sampling.

doi: 10.1007/s12016-026-09194-y

Mechanistic toxicology

3 records

Na HW et al. 2026 J Toxicol

Airborne metals reconstructed from worker sebum, then dosed onto skin

ICP-MS on sebum from **62 indoor and 40 outdoor workers** in Shanghai across 24 metals, with large outdoor–indoor contrasts (cobalt 925×, arsenic 135×, barium 29×, manganese 17×). Sebum-mimicking mixtures were then formulated at the measured profiles and applied to keratinocytes, sebocytes and ex vivo scalp skin: the outdoor mixture reduced viability, raised lipid peroxidation, induced γH2AX foci, upregulated *IL1A*, *IL1B*, *TNFA*, *IL6* and *PTGS2*, and activated NF-κB and MAPK. The design is the interesting part — a biomonitoring matrix used to specify the in vitro dose rather than an arbitrary slurry concentration. The weakness is that sebum metal load is a cumulative deposition proxy of unknown relation to inhaled or dermally absorbed dose, and indoor/outdoor is a crude exposure contrast with occupation confounded into it.

doi: 10.1155/jt/7925949

Wang Y et al. 2026 Environ Sci Technol Lett

α-Dicarbonyl aqueous products modulate cellular ROS in opposite directions

Glyoxal or methylglyoxal reacted with ammonium sulfate or glycine in the aqueous phase; HRMS shows ammonium routes yield imidazole-type nitrogen heterocycles and glycine routes yield more pyrrole-related species. Alone, neither product set induces much ROS in A549 cells. Co-exposed with 1,4-naphthoquinone, Cu²⁺ or both, **imidazole-dominated mixtures enhance ROS and pyrrole-rich mixtures suppress it**, with acellular assays attributing the split to promotion versus suppression of quinone redox cycling and metal-associated pathways. This is a direct challenge to any oxidative-potential assay read as a scalar: the same secondary organic aerosol aging pathway can raise or lower measured ROS depending on the nitrogen source. Cell-line work at unstated atmospheric relevance of concentration, so treat as mechanism, not as an exposure-response.

doi: 10.1021/acs.estlett.6c00722

Cao et al. 2026 Environ Sci Atmos

Black carbon interfacial chemistry and multiscale aging

A review that puts BC *particle state* — surface reactivity, coating thickness, morphology, aging timescale — rather than BC mass at the centre, and follows it through to optics, cloud activation, removal, lifetime and toxicity. The framing matters for instrumentation because every one of those downstream properties is a function of a variable that filter-based absorption and single-particle soot photometry constrain only partially, and that no low-cost optical sensor constrains at all. No primary data and no quantitative synthesis; the Crossref deposit carries only a graphical-abstract blurb, so the parameterisation recommendations were not verifiable at digest time. Listed as a reference framework rather than as evidence.

doi: 10.1039/d6ea00093b

Occupational & indoor

3 records

Srinivasan et al. 2026 Natl Med J India

Occupational wood dust and lung function — pooled, and heterogeneous

PRISMA review of PubMed and Google Scholar; 55 studies qualitatively, 16/14/9 pooled by outcome. Random-effects pooled mean differences in percent predicted: FEV₁ -8.57 (95% CI -12.07 to -5.08), FVC -7.43 (-11.35 to -3.51), PEFR -13.43 (-25.14 to -1.72). Sawmill workers trended obstructive, furniture workers restrictive. Three things limit it: **I² exceeds 95%** on every pooled outcome, the funnel plot indicates publication bias, and exposure is binary — no study contributed a dust concentration, so there is no exposure–response and no basis for an occupational limit. Searching only PubMed and Google Scholar is thin for an occupational literature that sits substantially in non-indexed regional journals.

doi: 10.25259/nmj1_668_2024

Feng R et al. 2026 Environ Sci Technol Lett

Clean heating renovation cuts nitro-PAHs but not nitrated phenols

Size-resolved *personal* exposure to particulate nitrated phenols (NPs) and nitro-PAHs among rural users of clean coal, raw coal chunk and biomass, paired with A549 cytotoxicity and NF- κ B reporter assays. Biomass users carried up to **11.5 $\times$** the PM_{2.5}-bound n-PAH exposure of clean-coal users, while NPs varied **≤ 1.3 -fold** across fuels (clean coal 64.0 ± 12.1 vs raw coal 57.4 ± 25.8 ng/m³, $p=0.545$). Toxicologically the two classes diverge as well: n-PAHs were more cytotoxic by IC₅₀, NPs activated NF- κ B more strongly via p-I κ B- α and COX-2. The policy reading is sharp — a fuel switch evaluated on PM mass or on PAHs alone will score a benefit that the NP fraction does not deliver. Group sizes are not stated in the abstract and the in vitro work uses representative compounds, not the collected particles.

doi: 10.1021/acs.estlett.6c00629

Jamali et al. 2026 Int J Biol Macromol

All-cellulose HEPA nanofilter in a wooden frame

Fog-sprayed, freeze-dried bacterial-cellulose nanofibres on a wood-derived cellulose-microfibre tissue substrate, in a modular wooden frame with a starch-adhesive gasket. Removal efficiency rises with BCNF grammage to **2 g/m²** and falls above it — consistent with pore closure raising velocity through the remaining paths. Optimum gives **99.998% PM_{0.3} at 150 Pa**, inside the EN 1822 H13–H14 band, and PM_{0.3} capture *peaks at high RH* under cyclic humidity, the opposite of the usual electret degradation. The claim being made is biodegradability at H13 performance, and the tests reported are single-pass efficiency and pressure drop; there is no loading lifetime, no microbial growth assessment on a cellulose substrate held at high RH, and no mechanical durability data. Promising material, incomplete filter.

doi: 10.1016/j.ijbiomac.2026.154178

Sensing, forecasting & instrumentation

11 records

Kaur et al. 2026 Aerosol Sci Technol

Plantower PMSX003N against FEM through a Utah winter

Three PMSX003N, three PMS5003T and three OPC-N3 co-located with FEM monitors at a Utah DEQ station for five months, plus laboratory size-binning against monodisperse DOS aerosol. The laboratory result drives everything: the PMSX003N **preferentially assigns particles $>1 \mu\text{m}$ to the 2.5–5 μm bin**. In dust events this recovers coarse mass its predecessor loses — **PM_{2.5} slope 1.85–1.91**, $R^2 > 0.7$ vs **0.15**, $R^2 < 0.37$; **PM₁₀ slope 0.91–1.07** vs **~ 0.02** — and under freezing fog, mist, snow and rain it substantially overestimates both fractions. Three units per model over one site-season is a small basis for a between-unit claim, and a slope near 1.9 is a large residual bias even where the correlation is good. Humidity-dependent correction is not optional for this part.

doi: 10.1080/02786826.2026.2716649

Guth et al. 2026 Atmosphere

Time-resolved emission factors for 19 prescribed burns, measured with low-cost sensors

Emission factors for CO, PM_{2.5}, EC, OC and speciated organics across **19 prescribed fires** in Colorado and southeastern Georgia, using multiple low-cost monitors per burn to get temporal resolution that filter-based EF campaigns cannot reach. CO EFs are driven by combustion phase — **+3260%** at low modified combustion efficiency, i.e. smouldering. Broadcast and pile burns are compared directly, which is rare. The methodological point for a distributed network is that the LCS bias that disqualifies a sensor for regulatory-equivalent concentration is tolerable here because the measurand is a ratio within a plume; the risk is that any RH- or size-dependent bias in the PM channel propagates straight into the PM_{2.5}/CO EF ratio, and the abstract does not report the sensor correction applied.

doi: 10.3390/atmos17090867

Xing et al. 2026 Environ Sci Technol

DeepAQM — observation-constrained ML surrogate for the NOAA forecast system

A ConvLSTM–ResNet fusion reconstructs spatially continuous surface PM_{2.5} and O₃ from VIIRS AOD, TEMPO NO₂ and EPA AirNow, and supplies them as initial conditions for 72 h UFS-AQM forecasts over the continental US. It reproduces UFS-AQM spatial patterns ($R^2 > 0.9$ for O₃, to 0.80 for PM_{2.5}, normalised mean bias within $\pm 10\%$). The result that matters: observation-updated initial conditions lift site-level R^2 in the first 10 forecast hours from **~ 0.10 – 0.15 to 0.40 – 0.50 for PM_{2.5}**, and observation-based fine-tuning carries overall R^2 to **0.41 for PM_{2.5}** across the full 72 h. Note what that number is: a fourfold gain at short lead, decaying to a forecast that still explains under half the variance. TEMPO makes this a current-generation result rather than a reanalysis exercise.

doi: 10.1021/acs.est.6c04600

Veselovskii et al. 2026 Atmos Meas Tech*Five-channel fluorescence lidar retrieves smoke mass inside the boundary layer*

Smoke and urban aerosol have distinct fluorescence spectra, and a multi-channel fluorescence lidar exploits that to separate them and retrieve smoke mass concentration with an estimated detection threshold of $\sim 0.1 \mu\text{g}/\text{m}^3$. Over Moscow, smoke was detected in **59 of 67 sessions between April and October 2024**, mostly from long-range Atlantic transport; only 12 episodes traced to southern Russian fires. Within the PBL, transported North American wildfire smoke contributed $\sim 1 \mu\text{g}/\text{m}^3$ against **up to $50 \mu\text{g}/\text{m}^3$** for regional fires. This is the source-resolved column-to-surface capability that mass-only networks lack, and $0.1 \mu\text{g}/\text{m}^3$ is a genuinely low threshold. Single site, single instrument, and mass retrieval requires an assumed fluorescence-to-mass conversion whose uncertainty is the main open term.

doi: [10.5194/amt-19-5587-2026](https://doi.org/10.5194/amt-19-5587-2026)**Zhang Z et al. 2026** Anal Chem*Microfluidic multispectral diffraction for black carbon quantification*

A microfluidic chip does inertial separation, low-velocity sheath-flow regulation and high-voltage electrostatic capture to enrich submicron BC in the imaging region (**92% enrichment efficiency**); a lensless diffraction system then acquires hyperspectral fingerprints over 400–800 nm, which a graph model maps to concentration. Over **$0.01\text{--}0.20 \text{ mg}/\text{m}^3$** , $R^2=0.98$. The discriminating claim is more interesting than the calibration: the spectral fingerprint separates BC from similarly-sized microplastics and pathogenic spores, which absorption-based instruments cannot do. Bench work only — no ambient co-location, no interference testing against real coated or aged BC, and $0.01 \text{ mg}/\text{m}^3$ ($10 \mu\text{g}/\text{m}^3$) is a high floor for ambient BC, which typically runs $0.5\text{--}5 \mu\text{g}/\text{m}^3$.

doi: [10.1021/acs.analchem.6c03051](https://doi.org/10.1021/acs.analchem.6c03051)**Giango et al. 2026** J Radiol Prot*Fifteen consumer radon monitors, 98 days, one occupied house*

45 units (15 models $\times$ 3) co-located with an AlphaGuard PQ2000 and two alpha-track detectors in the lowest lived-in space of an occupied Ontario home, with $\text{PM}_{2.5}$, CO_2 , temperature and RH logged alongside; reference mean **$99 \text{ Bq}/\text{m}^3$** (range 62–141). Cumulative performance: **8 of 15 models** had two or more units within $\pm 25\%$ relative percent error, the best reaching 3–5% after 13 weeks, while others ran from **–47% to +220%**. Intra-model CV was mostly under 15%; real-time RMSE for the best units was $5\text{--}6 \text{ Bq}/\text{m}^3$ with Spearman $\rho=0.70\text{--}0.77$ on two-day averages. Different analyte, and the QA design is directly transferable: this is the between-model spread estimate that consumer PM networks routinely lack. One home, one season, one concentration regime.

doi: [10.1088/1361-6498/aea16a](https://doi.org/10.1088/1361-6498/aea16a)**Cui C et al. 2026** Environ Sci Technol*Eighteen years of CCN reconstructed by XGBoost with SHAP attribution*

An observation-driven XGBoost framework reconstructs CCN number concentration at a North China Plain mountain site, 2007–2025. Particle number size distribution carries **$55.7\text{--}66.3\%$** of predictive importance, with the controlling size range shifting from 100–200 nm at 0.2–0.4% supersaturation to **$50\text{--}100 \text{ nm}$ at 1.0%** . No monotonic CCN trend, but summer bulk activation ratio and hygroscopicity parameter both **declined from 2009 to 2025**. Coupling PMF with the ML model attributes that to falling SO_2 influence and rising contributions from new particle formation and growth — a legible consequence of desulfurisation policy. Reconstruction over an 18-year record inherits every instrument change in that record, and SHAP importance is not a causal decomposition.

doi: [10.1021/acs.est.6c04501](https://doi.org/10.1021/acs.est.6c04501)**Galioto et al. 2026** Atmosphere*Physics-informed ML for street-canyon NO_2 in Palermo*

Six years of hourly data (2020–2025) on one of Europe's densest traffic corridors, resolved to **$\sim 11 \text{ m}$** : COPERT-Italy emission factors over the OpenStreetMap network, dispersed with the SIRANE street-network model, then fused with measurements in an ML stage and mapped in GIS. Observation–model scatter over **44,752 hours** centres on the 1:1 line. The genuinely useful move is inverting the chain to recover corridor traffic volume from observed NO_2 — a template for using a fixed monitor as a traffic sensor, which transfers to PM only where the non-exhaust fraction is separately handled. Analyte is gas-phase; single corridor; and a physics-informed hybrid validated against the same stations that trained it needs an out-of-corridor test before the inversion can be trusted.

doi: [10.3390/atmos17090865](https://doi.org/10.3390/atmos17090865)**Shafi & Scafetta 2026** Environ Sci Pollut Res Int*Reconstructing a decade of missing Delhi monitoring records*

Delhi went from **4 stations in 2014 to 45 in 2024**, leaving a fragmented record that defeats trend analysis. For each missing daily value across six pollutants, the workflow selects the four most-correlated stations as predictors, evaluates **35 regression models** in MATLAB's Regression Learner, takes the best, and iterates until the matrix is full. Tree ensembles and Gaussian/rational-quadratic kernels consistently beat multilinear regression; artificial-gap experiments validate across pollutants, stations and years, best for $\text{PM}_{2.5}$ and other regionally-driven species. The structural problem is unaddressed and important: stations are missing non-randomly, and imputing from correlated neighbours smooths exactly the local excursions that a network is deployed to detect. Reconstructed series should not be used for exceedance statistics.

doi: [10.1007/s11356-026-38181-1](https://doi.org/10.1007/s11356-026-38181-1)**Alderotti et al. 2026** Sci Total Environ*Raman spectroscopy as a non-destructive biomonitoring readout*

Mature *Quercus ilex* sampled along a high/intermediate/low NO_2 and PM_{10} gradient in Tuscany. Leaf Raman spectra show coordinated modulation of pigment (chlorophyll, carotenoid) and inducible flavonoid bands, cross-validated against destructive biochemistry and chlorophyll fluorescence, and consistent with *acclimation* rather than photoinhibitory damage. A composite Raman index tracked site-level NO_2 most closely, which the authors read as nitrogen driving the response. The attraction is a rapid, non-destructive, scalable field readout; the limitation is that a three-site gradient cannot separate NO_2 from PM_{10} from any co-varying urban factor, so the index is at best a composite urban-exposure indicator and not a pollutant-specific one.

doi: [10.1016/j.scitotenv.2026.182253](https://doi.org/10.1016/j.scitotenv.2026.182253)**He J et al. 2026** ACS Sens*Room-temperature ppb NO_2 chemiresistor by molecular packing control*

Alkyl chains of varying length on cyano-substituted distyrylbenzene tune molecular conformation and film microcrystallinity, exposing more surface-accessible functional groups and raising NO_2 sensitivity; the optimised device reaches a **30 ppb** detection limit at room temperature, and response is *enhanced* by humidity through $\text{NO}_2\text{--H}_2\text{O}$ synergy. Demonstrated in a flexible wireless vehicle-exhaust monitor. Carried here on the sensing axis with the analyte stated plainly: this measures a gas, not a particle, and contributes nothing to PM mass or size. It is included because room-temperature operation and positive rather than negative humidity dependence are both unusual for the co-located gas channels that sit in multi-pollutant low-cost nodes. No long-term drift or cross-sensitivity data.

doi: [10.1021/acssensors.6c00977](https://doi.org/10.1021/acssensors.6c00977)

Exposure assessment & modelling

12 records

Bravo et al. 2026 J Expo Sci Environ Epidemiol*Chicago's 2023 wildfire smoke, by tract*

Hourly PM_{2.5} on a 500 m grid aggregated to daily 24-hour means for all 792 Chicago census tracts in nine city regions, 1 May–31 July 2023, examined against demographic, socio-economic and health profiles. The title carries the finding: during the long-range wildfire-smoke episodes, the usual within-city exposure gradient compressed — smoke reached every region — while the pre-existing disparity in baseline PM_{2.5} and in underlying health persisted underneath it. That combination is the argument for source-resolved rather than total-mass exposure metrics in environmental justice work, since a total-mass summer average would show convergence and hide the differential burden. Modelled rather than measured exposure at tract level, one summer, one city, and no health outcome is linked here.

doi: 10.1038/s41370-026-00956-6

Hulme et al. 2026 Environ Res Lett*Home and school exposure, jointly, for 919 Accra schoolchildren*

Land-use regression annual means for PM_{2.5}, NO₂ and BC plus L_{day}, L_{night} and L_{den} noise, linked to geocoded home and school points for 919 of 1,034 children aged 8–12 across 90 Accra schools, then combined into a time-location-weighted total. Median home PM_{2.5} 33.8 µg/m³, NO₂ 55.5, BC 5.2 × 10⁻⁵ m⁻¹; school values slightly lower (29.9 / 53.5 / 4.9). Time-weighted totals exceeded WHO guidance for most children, and noise exceeded WHO road-traffic thresholds at home (L_{den} 66.3) and Ghana's educational-area standard at school. The joint air-plus-noise, home-plus-school design is the contribution — both are under-measured in sub-Saharan Africa. Annual LUR means cannot resolve the school day, so the time-weighting is coarser than it looks.

doi: 10.1088/1748-9326/ae8d40

Liu S et al. 2026 Environ Pollut*Wind-shear regimes and the vertical PM_{2.5} profile in a basin*

Six instrument types — surface network, wind profiler radar, microwave radiometer, aerosol lidar, radiosonde and UAV — plus WRF, over Taiyuan Basin. Unsupervised clustering gives three vertical-wind-shear regimes: strong directional near-surface shear with near-surface accumulation and the shallowest boundary layer; speed shear above 500 m with an elevated PM_{2.5} maximum near that height; and deep shear below 2000 m with stronger winds and low concentrations. UAV profiles over 0–500 m confirm the three shapes. The 17 January 2024 episode makes the operational point: strong shear did not enhance vertical exchange while the shear layer sat above a shallow stable layer, and only coupled once daytime heating deepened the boundary layer into it. Directly relevant to siting — a surface monitor cannot see regime 2 at all. One basin, one winter.

doi: 10.1016/j.envpol.2026.129103

Yu H et al. 2026 Environ Sci Technol*Sea-land breeze recirculation of black carbon in Qingdao*

A full year of online BC (2021) split into fossil-fuel and biomass-burning fractions by the Aethalometer model, then attributed to meteorology by random forest and trajectory analysis. Annual mean 1.78 ± 1.29 µg/m³; fossil fuel 88.2% of mass, biomass burning rising to 19% in winter. The two fractions respond differently — BC_{ff} to wind-driven ventilation and transport, BC_{bb} to low temperature — and BC enhancement under sea-land breeze peaks at 2.0–3.5 m/s, with diurnal persistence and directional shifts consistent with return transport and reaccumulation. Coastal siting guidance follows directly: a single coastal monitor under SLB conditions is sampling a recirculated air mass, not a fresh one. Aethalometer-model source splitting depends on assumed absorption Ångström exponents, which is the dominant uncertainty here.

doi: 10.1021/acs.est.6c10060

Wu Q et al. 2026 Environ Sci Technol*Nighttime new particle formation, and it is not always sulfuric acid*

Roughly 1.5 years of molecular-level measurement in coastal Dalian. Spring and winter nocturnal NPF is driven by H₂SO₄, whose dimer formation proceeds near the collision limit. In summer and warm autumn, H₂SO₄ and its dimers are scarce while oxygenated organic molecules are abundant, and OOM dimers correlate with nascent particles at mean $r=0.55$. Organic-driven nucleation has been reported for biogenic environments; showing it under anthropogenic influence at night is the novel claim. Implication for number-concentration monitoring: an ultrafine source term exists at night that photochemically-anchored parameterisations will miss. Correlation of 0.55 is suggestive rather than mechanistic, and one coastal site cannot establish how general the seasonal switch is.

doi: 10.1021/acs.est.6c04493

Gao J et al. 2026 Environ Sci Technol*Are inverse ozone–health associations an artefact of measuring the wrong oxidant?*

A perspective, not a study. Population studies not infrequently find lower O₃ associated with higher disease risk. NO titration near traffic and differential measurement error are the standard explanations and do not fully account for it. The alternative advanced here: low measured O₃ can mark locations and times of active oxidative chemistry — O₃ being consumed, or parallel oxidant pathways running — which generate toxic secondary products that no monitoring network measures. If so, O₃ is functioning as a negatively-signed proxy for an unmeasured oxidant field. The same logic transfers directly to PM mass as a proxy for oxidative potential, which is why this is carried in a PM watch. No data and no test proposed beyond a call for expanded measurement, so it stands or falls on plausibility.

doi: 10.1021/acs.est.5c17322

Yang & Li 2026 Environ Sci Technol*Modelling the whole indoor-radon distribution, not the area mean*

Over 6 million short-term radon measurements across the contiguous US, 2000–2021, used to predict percentiles at ZIP Code Tabulation Area level and then to model the within-ZCTA distribution rather than summarising each area by a mean or median. The consequence is quantitative: roughly half of US ZCTAs have at least one in five households above policy-relevant thresholds — a figure an area-average surface cannot produce, because the exceeding fraction is a property of the spread. Same argument applies to any PM exposure surface used for threshold-crossing statistics: modelling the mean and modelling the tail are different tasks. Radon test data are self-selected (homeowners who chose to test, often at point of sale), which biases both the level and the spread in unknown directions.

doi: 10.1021/acs.est.6c01333

Lai et al. 2026 Environ Sci Technol

Two-thirds of ambient organosulfates have no assigned source

Organosulfates contribute up to **30% of aerosol mass**. Of **366** ambient OS molecular formulas compiled from field studies, only about **36%** map to a known precursor or formation mechanism, leaving **64% unclassified**. The proposal here is that some unclassified species are not freshly formed at all but higher-generation products of heterogeneous and aqueous multiphase transformation of existing OS: identified transformation products already account for **~17%** of reported formulas and explain an additional **5%** of the unclassified set. Modest arithmetic, but it reframes a source-apportionment problem as an aging problem. Compilation-based, dependent on the sampling and HRMS coverage of the underlying field studies, and **5%** is a small demonstration of a mechanism claimed to be general.

doi: 10.1021/acs.est.6c08394

Schaefer et al. 2026 Atmos Chem Phys

Deep convection sorts ice-nucleating particles by activation temperature

Immersion-mode INP filter samples from the HALO aircraft up to **14.5 km** during CIRRUS-HL (summer 2021), taken in the inflow and outflow of deep convective clouds and from cloud-particle residuals. Inflow spectra contain INPs active at both high ($T > -15^{\circ}\text{C}$) and low ($T < -20^{\circ}\text{C}$) temperatures; **outflow spectra contain only the low-temperature population**. The reading is precipitation scavenging in the updraft removing the warm-active INPs while the cold-active ones are carried efficiently into the free troposphere. Cloud microphysics rather than exposure science, carried for the particle-transport result: convection is a size- and composition-selective vertical filter, which matters for any attempt to relate column-integrated aerosol to surface composition. Offline analysis, filter sampling, one campaign.

doi: 10.5194/acp-26-12505-2026

Wang X et al. 2026 Environ Monit Assess

Coupling coordination between bus-stop dust and soil metals

52 paired surface-dust and soil samples (104 total) at bus stops in Weiyang District, Xi'an, assayed for Cr, Mn, Cu, Zn, As and Pb and scored with Igeo and the Nemerow index. All but As and Mn exceeded Shaanxi soil background. Spatial patterns differ by medium — dust higher east, soil higher south — and sources resolve to traffic, natural and industrial. Hand-to-mouth ingestion dominated the exposure route and non-carcinogenic risk was higher for children than adults. The novel element is applying a coupling-coordination degree between the two media, which came out *moderate* and is read as partial interactive migration with shared sources. That index is a descriptive statistic borrowed from regional economics, not a transport model, so “interactive migration” is an interpretation rather than a demonstrated flux.

doi: 10.1007/s10661-026-15820-1

Pachés et al. 2026 Atmos Pollut Res

Regional background PM_{2.5} in the Valencian Community by clustering

Ships on metadata only. The Elsevier deposit carries no abstract, the record is not yet in PubMed, Europe PMC or Semantic Scholar, and ScienceDirect refused retrieval from this sandbox, so the network size, the clustering method and the resulting background estimates are all unavailable. Counted and listed; no numbers claimed. Flagged for the abstract retry list. Background/local-increment separation is directly relevant to siting a low-cost network — the increment is what a dense network is deployed to resolve — so this is worth revisiting once the abstract deposits.

doi: 10.1016/j.apr.2026.103192

Odu-Onikosi et al. 2026 Atmos Pollut Res

PM_{2.5}-bound toxic elements in Lagos

Ships on metadata only, same Elsevier deposit gap as above: no abstract, not in PubMed or Europe PMC, ScienceDirect blocked. Sampling duration, element list, apportionment method and risk estimates are all unretrievable at digest time. Counted and listed; no numbers claimed, and flagged for the abstract retry list. Noted because the author list includes P. A. Solomon and P. K. Hopke, and because speciated PM_{2.5} from Lagos — a megacity of roughly 15 million with almost no published speciation record — is exactly the geographic gap the two African records above document.

doi: 10.1016/j.apr.2026.103193

Other clinical endpoints

3 records

Batisse et al. 2026 Environ Epidemiol

Prenatal ultrafine particles and childhood cancer in 965,000 children

Cohort of over **965,000** children born in Montreal or Toronto 1999–2020, with high-resolution outdoor UFP number concentration and mean UFP size assigned to six-digit residential postal codes from annual tax records, followed to age 14 (mean 9.1 years) against the Canadian Cancer Registry. **1,625** incident cancers. Per 10,000 particles/cm³ of prenatal exposure: acute lymphoblastic leukaemia **HR 1.12 (0.94–1.34)**, neuroblastoma **1.15 (0.85–1.57)**; both intervals include the null, both persisted across exposure windows and sensitivity analyses, and other sites were null. Adjusting for UFP size as well as number is the methodological advance — number and size are correlated and separately toxicologically relevant. The honest summary is a consistent, imprecise positive signal, not a demonstrated association.

doi: 10.1097/ee9.0000000000000528

Chen K et al. 2026 Sichuan Da Xue Xue Bao Yi Xue Ban

PM_{2.5} components and incident frailty in CHARLS

CHARLS 2011–2020, $n=9,603$ middle-aged and older adults, frailty index outcome, exposures from the Tracking Air Pollution product, time-dependent Cox models. Per IQR of the preceding year: PM_{2.5} **HR 1.100 (1.015–1.193)**; by component, organic matter **1.142 (1.055–1.236)**, black carbon **1.131 (1.041–1.229)**, ammonium **1.098 (1.010–1.194)**, sulfate **1.088 (0.996–1.187)** and nitrate **1.081 (0.996–1.172)**. Carbonaceous components rank above bulk mass and above the secondary inorganic ions, and the two that cross the null are exactly the two secondary ions — an ordering consistent with combustion-source toxicity rather than with mass. The components co-vary strongly in the same exposure surface, so the ranking is suggestive and the six estimates are not six independent tests.

doi: 10.12182/20260760103

Park J et al. 2026 Int J Epidemiol*Ozone, disability and emergency department use in Korea*

Time-stratified case-crossover on **3,484,877** ED visits from the Korean National Health Insurance Service (2015–19, 2022–23), exposure the 3-day moving average of daily 8-hour maximum O₃, adjusted for daily mean temperature and holidays. Per IQR (~22 ppb): **+1.94% (1.64–2.25)** in persons with disabilities during the warm season and +1.93% (1.62–2.25) year-round, against **+1.36% (0.92–1.80)** in persons without. A roughly 40% larger relative effect in a population defined administratively rather than clinically, with an interaction term tested across temperature strata. No particulate metric is included, which is both the limitation and, read against Gao et al. in this issue, part of the point: a disability register can identify a susceptible group that pollutant-specific monitoring networks were never designed to serve.

doi: 10.1093/ije/dyag189

Neuro / mental health**1 record****Paulino et al. 2026** Int J Neurosci*Exposome burden and cognitive recovery after mild TBI*

44 adults with mild TBI (GCS 13–15, 50% female) prospectively enrolled at an academic clinic; 15 a priori exposome variables reduced by forward selection to 13 and summarised by PCA into a cumulative burden score, against residualised 6-month MoCA dichotomised into worse- and not-worse-than-expected recovery. The latent axis explains **16.0%** of variance and is loaded by education, area deprivation, insurance coverage, **PM_{2.5}** and adverse childhood events; burden was higher in the worse-than-expected group (median PC1 1.34 vs –0.76, $p=0.010$, holding under permutation). Carried for the **PM_{2.5}** loading only. At $n=44$ with 13 variables and an outcome-informed selection step, this is hypothesis-generating; the loading structure is mostly socioeconomic and **PM_{2.5}** may be riding on it.

doi: 10.1080/00207454.2026.2728613

Burden, policy & mitigation**1 record****Liu Z et al. 2026** Environ Sci Technol*Health co-benefits of China's 2035 NDC arrive before the climate benefits do*

Integrated assessment over 2025–2060 puts the NDC's net benefit at **+1,500.5 billion USD**, decomposing into **3,924.3 billion** in long-run climate benefits and **1,659.5 billion** in health co-benefits. The finding is the timing and the distribution, not the total: in the near term the NDC *reduces* climate benefit, especially in underdeveloped regions, because coal generation balances renewable intermittency and rapid afforestation releases soil organic carbon; long-run benefits reach all regions but stay concentrated in high-income ones. Air-quality health co-benefits are the component that accrues locally and immediately, which is the argument for treating PM monitoring as mitigation-policy infrastructure. Monetised co-benefits inherit every assumption in the concentration–response and valuation chain and should be read as scenario contrast, not as a forecast.

doi: 10.1021/acs.est.6c06024

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Paulino et al.	Neural exposome burden stratifies cognitive trajectories in a subacute-to-chronic mild TBI outpatient cohort: an exploratory dimension-reduction study	<i>Int J Neurosci</i>	Prospective cohort + PCA	Residential PM _{2.5} as one of 13 exposome variables entering a principal-component burden score, alongside education, area deprivation and adverse childhood events	North America
Reproductive & developmental					
Chen Y et al.	The impact of PM _{2.5} component exposure during early pregnancy on congenital heart disease: evidence from Guangdong, China	<i>Sci Rep</i>	Case-control (registry)	Satellite-derived PM _{2.5} components (OM, sulfate, nitrate, ammonium, BC) by week of early pregnancy; quantile g-computation for mixture weights	China
Xu ZL et al.	Timing matters: component-specific PM _{2.5} exposure windows and oocyte development in assisted reproduction	<i>Biomed Environ Sci</i>	Retrospective cohort + DLNM	Individual PM _{2.5} and its sulfate, nitrate, ammonium, organic matter and black carbon fractions over recent (0-3 mo), distal (4-12 mo) and cumulative (0-12 mo) windows	China
Respiratory & allergic					
Cui L et al.	Epidemic thunderstorm asthma in China: a systematic review of an Artemisia-associated phenotype	<i>Clin Rev Allergy Immunol</i>	Systematic review / meta-analysis	Airborne pollen as the causal particle; most Chinese reports identify sensitisation only to genus-level Artemisia, without species-level aerobiological confirmation	China
Wright et al.	Air pollution and heat impacts on respiratory morbidity and mortality outcomes in Africa: a systematic review towards a meta-analysis	<i>Curr Environ Health Rep</i>	Systematic review / meta-analysis	Reported indoor and ambient PM _{2.5} and nitrogen oxides across included case-control studies; no pooled exposure-response	Sub-Saharan Africa
Mechanistic toxicology					
Cao et al.	Interfacial chemistry and multiscale aging of atmospheric black carbon: from surface reactions to coating, morphology, and model parameterization	<i>Environ Sci Atmos</i>	Narrative / critical review	Black carbon particle state - surface reactivity, coating thickness, morphology and aging timescale - as the link between emission and optical, cloud and toxicological behaviour	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
Na HW et al.	Adverse effects of airborne heavy metals on human skin cells: insights from sebum analysis of indoor and outdoor workers	<i>J Toxicol</i>	Biomonitoring + in vitro	24 metals quantified by ICP-MS in sebum, then reconstituted as sebum-mimicking metal mixtures at the measured indoor and outdoor profiles	China
Wang Y et al.	Contrasting cellular reactive oxygen species modulation by alpha-dicarbonyl aqueous reaction products via interactions with redox-active particulate matter	<i>Environ Sci Technol Lett</i>	Bench / device experiment	Aqueous reaction products of glyoxal or methylglyoxal with ammonium sulfate or glycine, characterised by HRMS, then co-exposed with redox-active PM surrogates	Laboratory
Occupational & indoor					
Feng R et al.	Clean heating renovation differentially reduces nitro-PAH and nitrated phenol exposure: implications from real-world personal monitoring and NF-kB-based toxicity	<i>Environ Sci Technol Lett</i>	Personal exposure monitoring + in vitro	Size-resolved personal exposure to particulate nitrated phenols and nitro-PAHs among users of clean coal, raw coal chunk and biomass	China
Jamali et al.	Wooden-framed all-cellulose HEPA nanofilter for sustainable air purification	<i>Int J Biol Macromol</i>	Laboratory instrument development	PM0.3 removal efficiency and pressure drop of a fog-sprayed, freeze-dried bacterial-cellulose nanofibre layer on a cellulose-microfibre substrate	Laboratory
Srinivasan et al.	Occupational wood dust exposure and lung function impairment: a systematic review and meta-analysis	<i>Natl Med J India</i>	Systematic review / meta-analysis	Occupational wood dust exposure, categorical (exposed vs unexposed); no quantitative dust concentration pooled	Global / multi-region
Sensing, forecasting & instrumentation					
Alderotti et al.	Raman spectroscopy resolves nitrogen-driven metabolic acclimation to urban air pollution in <i>Quercus ilex</i>	<i>Sci Total Environ</i>	Bench / device experiment	Leaf Raman band intensities (chlorophyll, carotenoid, flavonoid) as a non-destructive biomonitoring index along an NO ₂ and PM ₁₀ gradient	Europe
Cui C et al.	Explainable machine learning prediction of long-term variations in cloud condensation nuclei at a high mountain site in the North China Plain	<i>Environ Sci Technol</i>	Data-fusion / ML surrogate model	CCN number concentration reconstructed from particle number size distribution and chemistry; bulk activation ratio and hygroscopicity parameter	China
Galioto et al.	Physics-informed machine learning for urban nitrogen dioxide forecasting in Palermo, Italy	<i>Atmosphere</i>	Chemical transport modelling	Hourly NO ₂ at 11 m resolution: COPERT-Italy emission factors dispersed through the SIRANE street-network model, then fused with measurements in an ML stage	Europe

First author	Focus	Journal	Design	Metric	Region
Giang et al.	Cumulative and real-time performance of consumer-grade electronic radon monitors in a Canadian single-family home over three months	<i>J Radiol Prot</i>	Sensor co-location / calibration	Radon activity concentration from 45 consumer monitors against an AlphaGuard PQ2000 and two alpha-track detectors; PM _{2.5} , CO ₂ , T and RH co-recorded	North America
Guth et al.	Exploring the variability in emissions for broadcast and pile prescribed burns using low-cost sensors	<i>Atmosphere</i>	Field emission-factor campaign	Low-cost-sensor emission factors for CO and PM _{2.5} , with filter-based EC/OC and speciated organics, resolved by modified combustion efficiency	North America
He J et al.	Molecular packing engineering enables a high performance ppb-level room-temperature NO ₂ sensor	<i>ACS Sens</i>	Laboratory instrument development	Gas-phase NO ₂ (not particulate): chemiresistive response of cyano-substituted distyrylbenzene films with tuned alkyl-chain length and molecular packing	Laboratory
Kaur et al.	Performance evaluation of Plantower's PMSX003N: laboratory and field analysis	<i>Aerosol Sci Technol</i>	Sensor co-location / calibration	PM _{2.5} and PM _{1.0} from three PMSX003N, three PMS5003T and three Alphasense OPC-N3 against FEM reference; laboratory size-bin response to monodisperse DOS aerosol	North America
Shafi & Scafetta	Management of missing air pollution data within urban environments using machine learning regressions: a case study for Delhi, India	<i>Environ Sci Pollut Res Int</i>	Data-fusion / ML surrogate model	Iterative reconstruction of missing daily PM _{2.5} , PM _{1.0} , O ₃ , NO ₂ , SO ₂ and CO from the four most-correlated neighbouring stations; 35 candidate regressors per gap	South Asia
Veselovskii et al.	Evaluation of smoke mass concentration within the PBL based on observations of fluorescence lidar with several discrete channels	<i>Atmos Meas Tech</i>	Remote sensing / lidar retrieval	Smoke mass concentration retrieved from five-channel fluorescence lidar spectra, separating smoke from urban aerosol; detection threshold 0.1 µg/m ³	Europe
Xing et al.	Multisource observation-constrained tuning for air quality forecasting in a machine learning-enhanced NOAA Unified Forecast System	<i>Environ Sci Technol</i>	Data-fusion / ML surrogate model	Surface PM _{2.5} and O ₃ fields reconstructed from VIIRS AOD, TEMPO NO ₂ and EPA AirNow, used as initial conditions for 72 h UFS-AQM forecasts	North America

First author	Focus	Journal	Design	Metric	Region
Zhang Z et al.	A microfluidic multispectral diffraction platform for quantitative analysis of atmospheric black carbon aerosols	<i>Anal Chem</i>	Laboratory instrument development	Black carbon mass concentration from lensless multi-spectral (400-800 nm) diffraction fingerprints after microfluidic inertial separation and electrostatic capture	Laboratory
Exposure assessment & modelling					
Bravo et al.	No one is immune: neighborhood trends and disparities in air pollution exposure in the city of Chicago during summer 2023 wildfires	<i>J Expo Sci Environ Epidemiol</i>	Exposure model (secondary data)	Daily 24 h tract-level PM _{2.5} from hourly 500 m gridded estimates, split by wildfire-smoke and non-smoke days	North America
Gao J et al.	Understanding unexpected ozone-health associations: are secondary oxidative pollutants key?	<i>Environ Sci Technol</i>	Narrative / critical review	Measured O ₃ as a proxy for an unmeasured oxidant field; argues low O ₃ can coincide with active oxidative chemistry producing toxic unmeasured secondary products	Global / multi-region
Hulme et al.	Exposure of children to multiple long-term air pollutants (PM _{2.5} , BC, NO ₂) and noise at home and school in Accra, Ghana	<i>Environ Res Lett</i>	Exposure model (secondary data)	Annual mean PM _{2.5} , NO ₂ and BC from land-use regression at geocoded home and school points, combined into a time-location-weighted total; Lday/Lnight/Lden noise alongside	Sub-Saharan Africa
Lai et al.	What are the sources of unclassified organosulfates in the atmosphere?	<i>Environ Sci Technol</i>	Narrative / critical review	Molecular formulas of ambient particulate organosulfates, which contribute up to 30% of aerosol mass, mapped to known precursors and formation routes	Global / multi-region
Liu S et al.	Wind shear structures and their association with the vertical distribution of PM _{2.5} in Taiyuan Basin	<i>Environ Pollut</i>	Multi-platform observation synthesis	Vertical PM _{2.5} profiles from aerosol lidar and UAV, against vertical wind shear regimes from wind profiler radar, microwave radiometer, radiosonde and WRF	China
Odu-Onikosi et al.	Characteristics, sources, and health risks of PM _{2.5} -bound potentially toxic elements (PTEs) in Lagos, Nigeria	<i>Atmos Pollut Res</i>	Metadata only	PM _{2.5} -bound potentially toxic elements with source apportionment and health-risk characterisation	Sub-Saharan Africa
Paches et al.	Estimation of background PM _{2.5} concentrations in the Valencian Community applying clustering technique	<i>Atmos Pollut Res</i>	Metadata only	Regional background PM _{2.5} separated from local increment by clustering of monitoring-network records	Europe

First author	Focus	Journal	Design	Metric	Region
Schaefer et al.	Upward transport and segregation of ice-nucleating particles in deep convective clouds	<i>Atmos Chem Phys</i>	Aircraft measurement campaign	Immersion-mode INP concentration-temperature spectra from filter samples of convective inflow, outflow and cloud-particle residuals	Europe
Wang X et al.	Interactive migration of heavy metals between dust and soil demonstrated by the coupling coordination analysis	<i>Environ Monit Assess</i>	Source-apportionment method	Cr, Mn, Cu, Zn, As and Pb in paired surface dust and soil, scored with Igeo and the Nemerow index and coupled across the two media	China
Wu Q et al.	Molecular characterization of nighttime new particle formation in a coastal urban environment: season-dependent roles of sulfuric acid and oxygenated organics	<i>Environ Sci Technol</i>	Long-term field measurement campaign	Nocturnal H ₂ SO ₄ , its dimers and oxygenated organic molecules measured alongside nascent particle number, by nitrate- and Vocus-type mass spectrometry	China
Yang & Li	From percentiles to distributions: nationwide ZCTA-level modeling of indoor radon exposure across the contiguous United States, 2000-2021	<i>Environ Sci Technol</i>	Exposure model (secondary data)	Within-area distributions of indoor radon concentration modelled at ZCTA level, rather than area means or medians	North America
Yu H et al.	Machine learning attribution of atmospheric black carbon in a coastal city: source variability and the sea-land breeze recirculation mechanism	<i>Environ Sci Technol</i>	Source-apportionment method	Online BC split into fossil-fuel and biomass-burning fractions by the Aethalometer model, then attributed to meteorology by random forest and trajectory analysis	China
Burden, policy & mitigation					
Liu Z et al.	Significant near-term inequity under China's 2035 NDC due to a time lag in mitigation effects across regions	<i>Environ Sci Technol</i>	Integrated assessment modelling	Air-quality health co-benefits monetised alongside climate benefits under the 2035 NDC pathway, resolved by region and by decade	China
Other clinical endpoints					
Batisse et al.	Exposure to outdoor ultrafine particle number concentrations and incidence of childhood cancer: a population-based cohort study	<i>Environ Epidemiol</i>	Prospective cohort (registry linkage)	High-resolution outdoor UFP number concentration and mean UFP size at six-digit residential postal codes from annual tax records; prenatal and postnatal windows	North America

First author	Focus	Journal	Design	Metric	Region
Chen K et al.	Effect of long-term exposure to PM _{2.5} and its chemical components on frailty incidence in middle-aged and older adults in China: a prospective cohort study	<i>Sichuan Da Xue Xue Bao Yi Xue Ban</i>	Prospective cohort (registry linkage)	One-year mean PM _{2.5} and its black carbon, organic matter, sulfate, nitrate and ammonium fractions from the Tracking Air Pollution dataset, per IQR	China
Park J et al.	Association between ozone and emergency department visits in people with disabilities in South Korea: a nationwide case-crossover study	<i>Int J Epidemiol</i>	Time-stratified case-crossover	Daily 8 h maximum ozone, 3-day moving average (lag 0-2), per IQR (22 ppb), adjusted for daily mean temperature; no particulate metric	South Korea

Provenance and method

Entry window **3 September 2026**, single day, opened against a `last_entry_date` of 2 September. Sources queried on 4 September 2026 at 11:3x CDT; under this pipeline's 22:00-local rule, 3 September is the newest buildable date and 4 September is not opened. No day is owed: 1 and 2 September shipped on the previous run. **PubMed** E-utilities on [EDAT] 2026/09/03: **29** hits on the health axis, **6** on the instrumentation axis. **Europe PMC** `CREATION_DATE`: 23. **Crossref** by-ISSN, windowed on `created`: **94** deposits. **arXiv** answered normally this run (60 entries, no HTTP 429, the first clean response since 31 August) but its newest submission is 2026-08-20, so the undated relevance query again contributed nothing; **OpenAlex** returned HTTP 429 (the plan-gated date filter has been broken since 28 July). **Consensus** returned ten low-cost-sensor calibration papers, every one published 2018–2024 and therefore out of window — the fifth consecutive nil run for that leg. Full text was pulled through the browser for two records only: the Plantower evaluation (Taylor & Francis, JavaScript-gated) and the Guangdong CHD paper, whose PubMed abstract renders its strongest component as “[Formula: see text]” and which was resolved to *sulfate* at source rather than guessed. ScienceDirect refused retrieval, which is why the two *Atmos. Pollut. Res.* records ship on metadata only. After collapsing cross-source duplicates by PMID, DOI and normalised title, **138 unique records** were considered; **12 were already shipped** in earlier issues, **38 were screened in scope** and **88 were screened out**. Notable exclusions: a tire-wear-particle earthworm toxicology paper and a Cu/SSZ-13 SCR catalyst study, both airborne-adjacent but with no ambient PM metric or human endpoint; two *Eur. J. Public Health* conference abstracts on indoor air quality in cystic fibrosis and the UKHSA air-quality dashboard, both on topic but with no retrievable text; and a *Health Aff. Sch.* commentary on weather-resilient primary-care payment, which carries no exposure metric. **Trial watch**: zero registrations or updates for this date across five air-pollution query terms. This is a true zero and not a query failure — the same date filter with no term returns 1,001 studies updated 3 September. Logged to `state/trials.json` for the W36 weekly. Two DOIs (10.12182/20260760103, 10.3967/bes2026.072) are registered outside Crossref and were initially failed as dead by the DOI gate; both resolve in the Handle system and the gate was extended accordingly, so both links are live. Subtopic, design, particle-metric and geography labels are single-label by dominant contribution and are a judgement, not a controlled vocabulary. All DOI links resolve to the publisher of record; abstracts remain the copyright of their respective publishers.